Introduction

Image segmentation is an essential step in many quantitative imaging workflows, enabling anatomical regions to be isolated for visualisation, volumetric measurement, and further analysis. In this tutorial, we demonstrate a semi-automated lung segmentation workflow using 3D Slicer, a free and open-source medical image analysis platform.

The example dataset was acquired on the 100 mT very low-field MRI scanner located at the AMT Center, University of Aberdeen. The data have an isotropic spatial resolution of 3.8 mm × 3.8 mm × 3.8 mm and provide a useful example of the challenges associated with emerging imaging technologies, including lower spatial resolution and image contrast characteristics that differ from those of conventional clinical scanners.

1. Senn N, et al. (2025) Feasibility of self-navigated motion resolved 3D lung MRI on a non-commercial 100 mT system. ESMRMB 2025. https://doi.org/10.1007/s10334-025-01278-8 no. 461.

2. Senn N, et al. (2026) Pulmonary ventilation mapping using a non-commercial very-low field MRI system. ISMRM 2026. http://echo.ismrm.org/abstracts/view/739c0d55-ae32-4416-8bef-d2695f4b85ac. no. 302-02-004

3. Senn N, et al. (2026) Free-breathing 4D lung imaging using a non-commercial very-low field MRI system. ISMRM 2026. http://echo.ismrm.org/abstracts/view/64e4754f-be0c-4fea-a398-685a88fee8fb no. 560-05-006.

Rather than relying on highly specialised segmentation methods developed for a specific imaging modality, this tutorial introduces the Grow from Seeds approach within 3D Slicer. This method combines limited manual annotation with automated region growing and is well suited to novel or unconventional datasets where standard segmentation tools may not perform reliably. The workflow is straightforward to learn, requires only a small amount of manual input, and can be readily adapted to a wide range of thoracic imaging studies.

Objective

The objective of this tutorial is to introduce users to a robust and accessible workflow for whole-lung segmentation in 3D Slicer.

By the end of this tutorial, participants will be able to:

- Create segmentation labels for the left lung, right lung, and surrounding tissues.
- Use the Paint tool to define representative seed regions in multiple imaging planes.
- Initialise and refine a Grow from Seeds segmentation.
- Assess segmentation quality using multiplanar and 3D visualisation.
- Save and export the completed segmentation for subsequent quantitative analysis.

This workflow provides a practical starting point for researchers and clinicians who are new to medical image segmentation and wish to analyse lung datasets acquired using either conventional or emerging imaging technologies.

Summary of Steps

- Create separate segments for the left lung, right lung, and other tissues.
- Draw only representative seed regions rather than complete contours.
- Annotate at least two slices in each imaging plane.
- Use an iterative cycle of Paint → Initialize → Review → Correct.
- Finalise the segmentation only after reviewing the volume in all three planes and in 3D.

Step 0.1: Download 3D Slicer

Step 0.2: Download the example data from:

https://github.com/nicholas-senn/teaching-resources/raw/refs/heads/main/lung-MRI/Advanced-Chest-MRI-School-2026/Lung%20Segmentation%203DSlicer%20First%20Steps/Lung_D1.nii.gz

Step 1: Load the Image Volume

1. Open **3D Slicer**.
2. Select **File → Add Data** and load the lung image volume.
3. Verify that the dataset is displayed in the axial, sagittal, and coronal views.

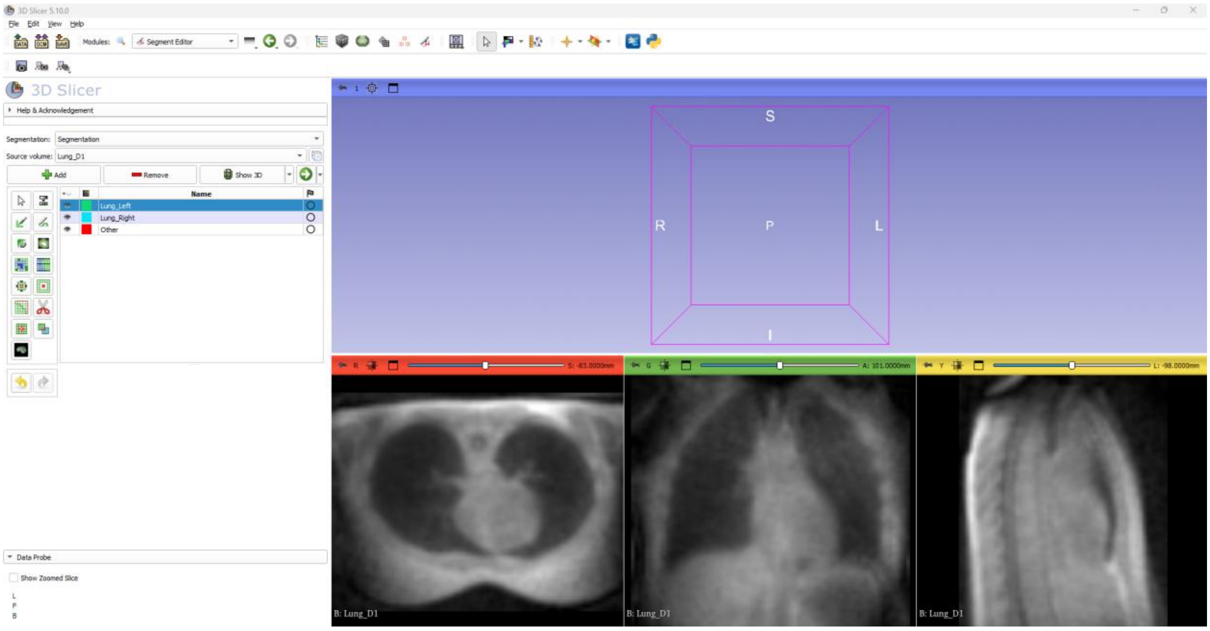

Figure 1. Step1. Data load successfully. Step 2. Three segments defined.

Step 2: Open Segment Editor

1. Open the **Segment Editor** module.
2. Create three segments by clicking **Add**:
    - **Left Lung**
    - **Right Lung**
    - **Other Regions**

The "Other Regions" segment is important because it provides the algorithm with examples of tissues that should not be included in the lung segmentation.

See Figure 1.

Step 3: Paint Seed Regions

Select the **Paint** tool and begin drawing representative regions.

Draw several strokes within the left lung, right lung on at least:

- Two axial slices
- Two sagittal slices
- Two coronal slices

Other Regions

On the same slices draw representative regions outside the lungs, for example:

- Chest wall
- Soft tissue
- Background

Again, annotate at least two slices in each imaging plane.

- Make sure you provide example drawing in coronal plane of the trachea and primary bronchi.

The goal is not to outline the entire anatomy. Instead, provide representative examples of each region so that the algorithm can classify similar voxels throughout the volume.

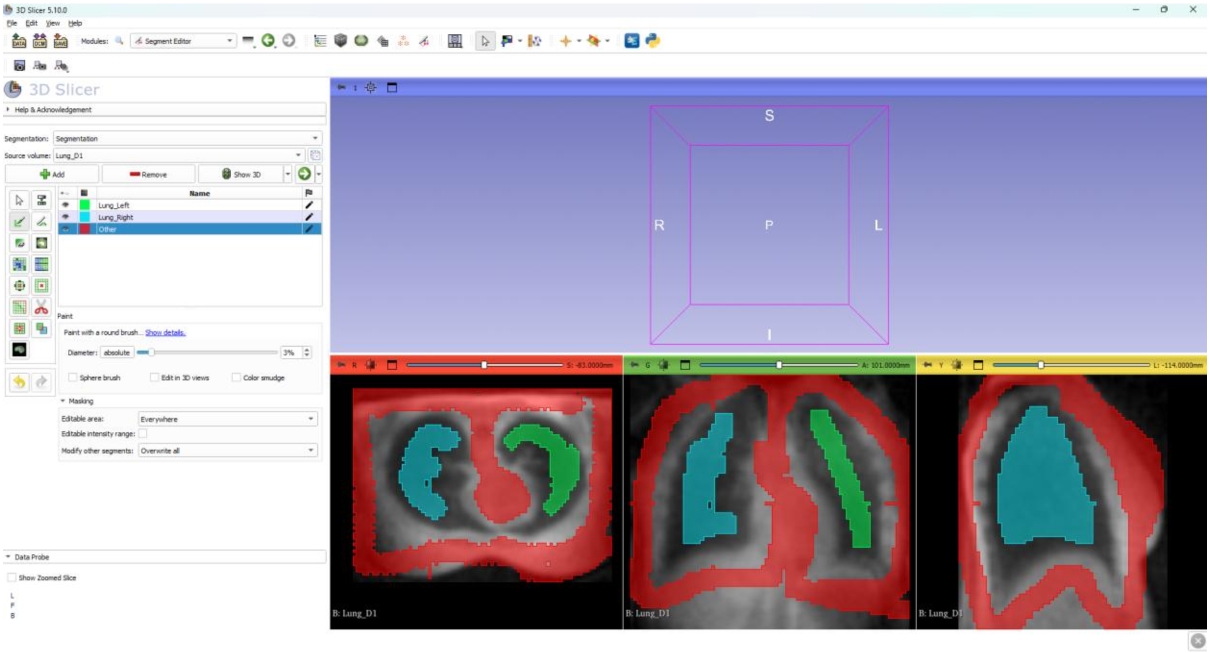

Figure 2. Step 3: Paint Seed Regions

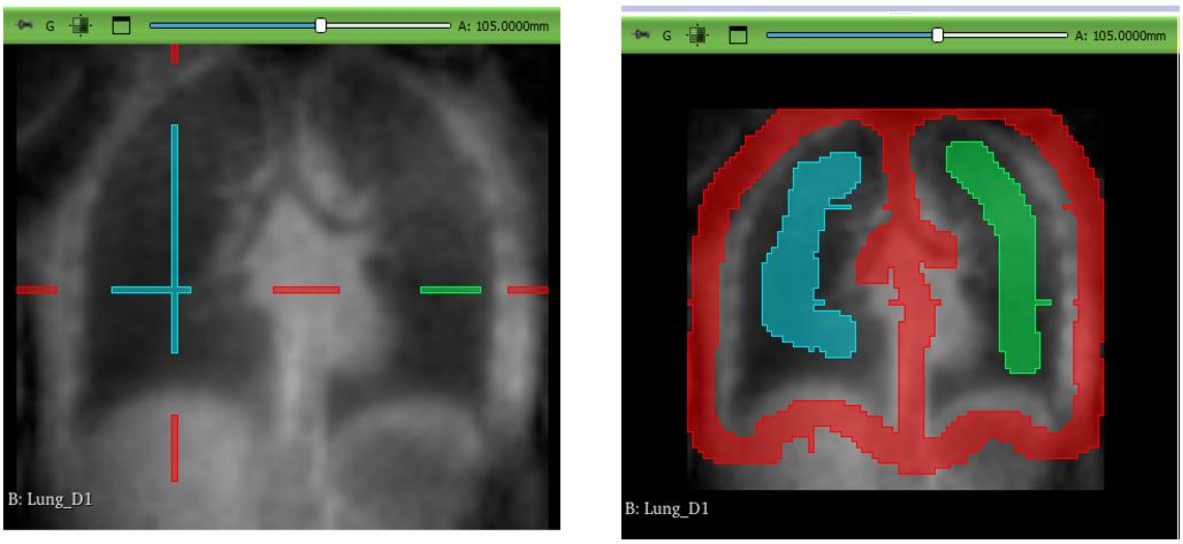

Figure 3. Step 3. Drawing regions of trachea and primary bronchi

Step 4: Initialise Grow from Seeds

1. Select **Grow from Seeds** from the Segment Editor effects list.
2. Click **Initialize**.

3D Slicer will generate a preview segmentation based on the seed regions that have been provided.

3. Click **Show 3D**

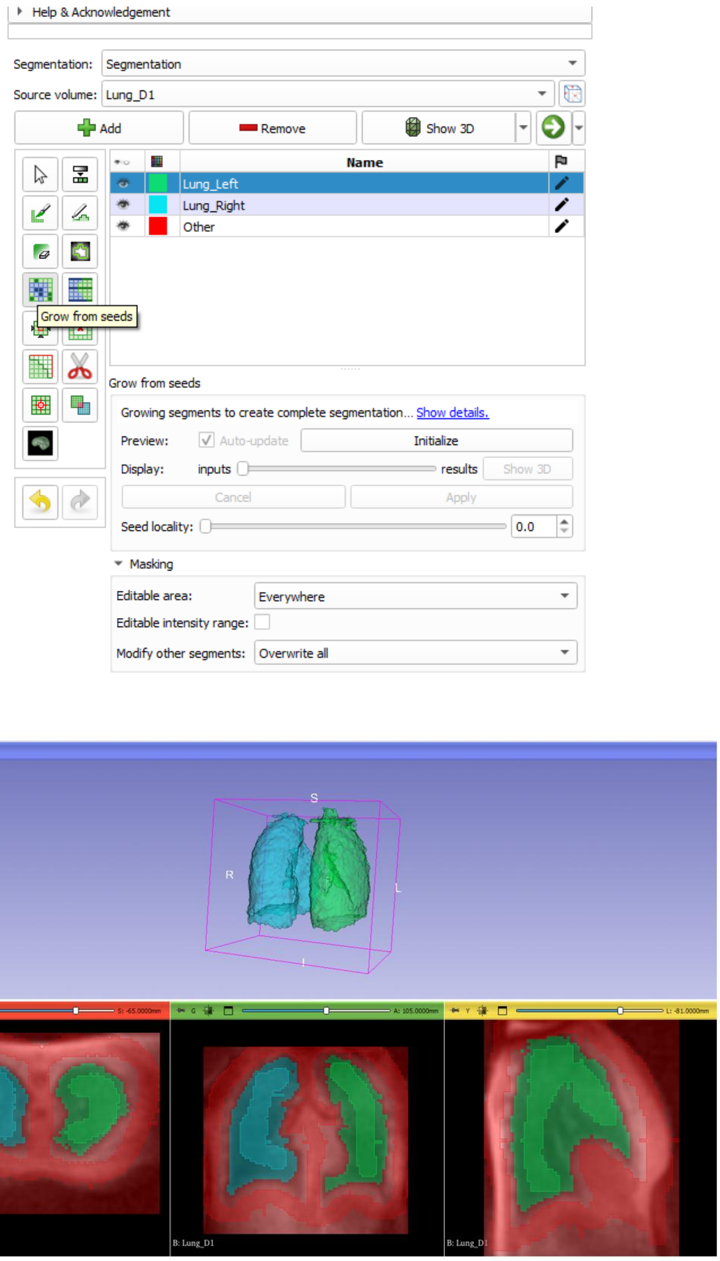

Figure 4. Step 4. Initialise Grow from Seeds.

Step 5: Review the Preview Segmentation

Scroll through the volume in all three planes and assess the segmentation.

Pay particular attention to:

- Lung boundaries
- Diaphragm interfaces
- Areas affected by image artefacts

The coloured preview should provide a reasonable representation of the left and right lungs, but correction is usually required.

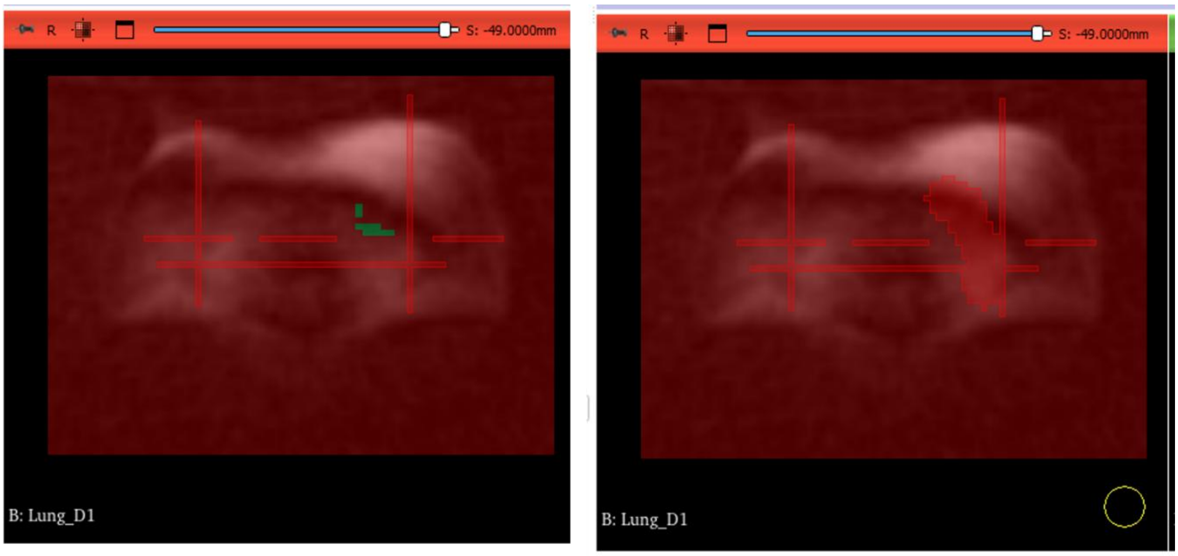

Figure 5. Steps 5 and 6: reviewing the preliminary segmentation and refining regions using the pen tool.

Step 6: Refine the Seed Regions

If parts of the lungs are incorrectly classified:

1. Return to the **Paint** tool.
2. Add additional strokes to the appropriate segment.
    - Paint into missing lung regions if lung tissue is excluded.
    - Paint into "Other Regions" where non-lung tissue has been included.
3. If Auto-update is ticked, the segmentation result will update after each drawing

This iterative process allows the segmentation to improve progressively with relatively little manual input.

Step 7: Apply the Segmentation

Once the preview is satisfactory:

1. Click **Apply**.
2. The preview segmentation will be converted into the final segmentation.

You can now inspect the result in all three imaging planes.

More tools and processing are available in Segmentation Editor.

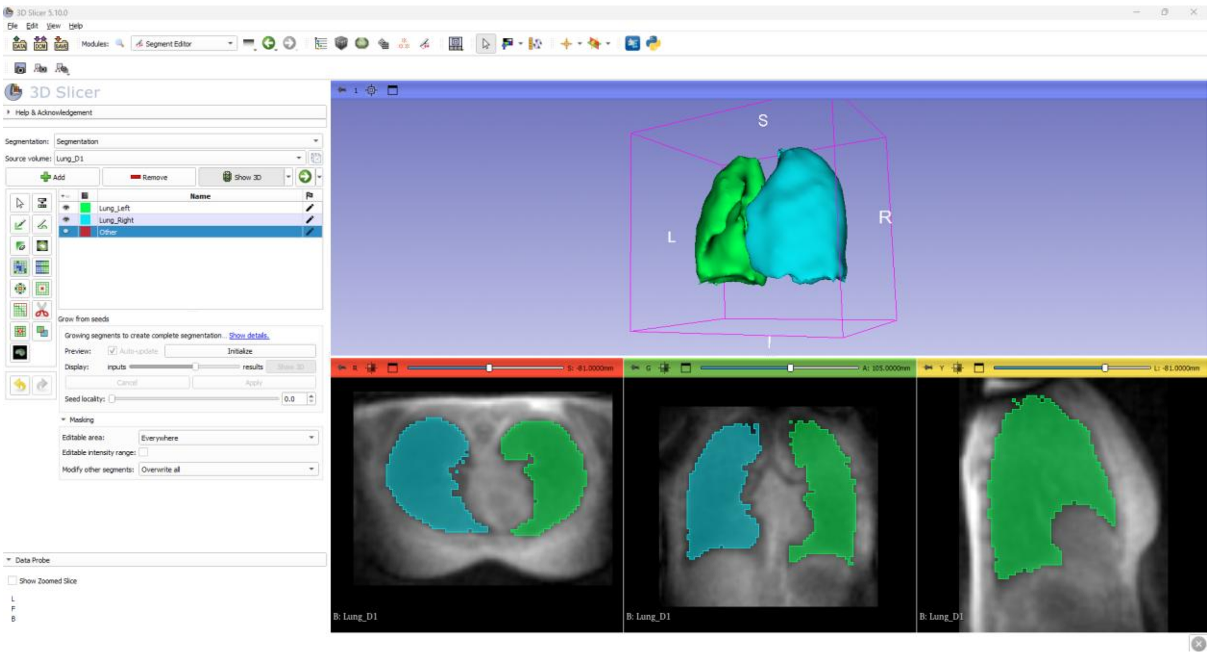

Figure 6. Step 7: Applied segmentation.

Step 8: Save the Segmentation

1. Select **File → Save**.
2. Save the segmentation as a **.seg.nrrd** file.
3. Export as a label map or surface model if required for subsequent analysis.

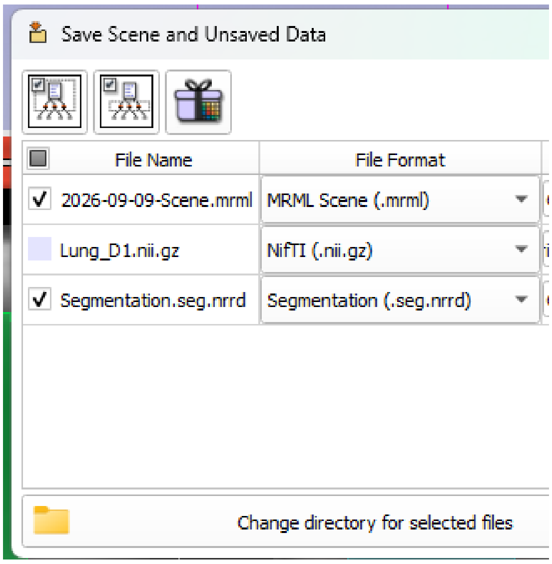

Figure 7. Step 8: Saving the segmentation result.

Final comments

There are many methods to create segmentation masks. This method has been shown because it is robust and works well for the novel imaging data set acquired at very low field. However:

- How long did the segmentation task take you?
- How much time was needed to correct the mask?
- How did you decide what level of accuracy you wanted?
- If you want to develop a full automated segmentation method using deep learning AI, how many training and testing data sets would you want?

Looking at this tutorial during the Chest MRI School?

Please remember to add details of any segmentation tools you might recommend to the share board.

The share recommended tools and resources will be uploading to the GITHUB page.

I end by saying thank you to Dr Gabriel Zihlmann for first introducing me to the 3D Slicer Segmentation Editor tool.